PEER-REVIEW AGENT

A complete worked-conversation teaching edition: manuscript review, 28-dimension scoring, revision triage, construct renaming, and decisions about collecting new data.

Teaching case: Heritage-product consumer evaluation manuscript | Prepared 12 October 2026

> Purpose. Show researchers how to use a rigorous peer-review agent iteratively: first establish a defensible audit, then translate reviewer criticism into verified revision actions, and finally decide whether construct relabeling or additional data are actually warranted.

Source and fidelity notice. This document reproduces the substantive conversation as an organized instructional transcript and provides reusable prompts, scoring matrices, and explanatory annotations. It is a teaching edition, not a platform-generated, byte-for-byte chat export. The manuscript itself was not appended in full. The uploaded Beihang-Referee software archive was not executed or benchmarked in this conversation. Page and line anchors below refer to the 34-page manuscript reviewed in the conversation.

Contents

1. Case context and what was submitted
2. Conversation 1: full reviewer instruction and editorial verdict
3. Scoring rubric: 28 categories
4. Construct-by-construct review and major criticisms
5. Methodology, statistics, fsQCA, and references
6. Revision checklist and journal decision
7. Conversation 2: Do we need new data or renaming?
8. Conversation 3: Is renaming acceptable for adapted items?
9. Practical workflow and reusable prompts
10. Integrity, limitations, and teaching exercises

1. Case context and uploaded evidence

The researcher submitted an empirical manuscript entitled Continuity, Credibility, and Contemporary Relevance in Heritage Product Evaluation: Traditional Craftsmanship Appraisal, Perceived Cultural Authenticity, and Perceived Contemporary Relevance, together with a Beihang-Referee ZIP archive. The manuscript covered 34 pages and reported an online quota-panel survey of 496 eligible Pakistani adults who had bought heritage products during the preceding 12 months.

The specified model related Traditional Craftsmanship Appraisal (TCA), Perceived Cultural Authenticity (PCA), and Perceived Contemporary Relevance (PCR) to Perceived Cultural Continuity (PCC), then to Positive Affective Response (PAR) and Buying Intent (BI). Product Knowledge (PK) and Traditional Cultural Orientation (TCO) moderated designated structural paths. The analysis used PLS-SEM and fsQCA.

The researcher asked not for editing first, but for a senior-journal-reviewer evaluation with passage-level anchoring, scientific critique, and numerical scores across more than 20 categories.

2. Conversation 1 - Review request and answer

User instruction: the original review rubric

> “You are acting as a senior peer reviewer for a high-level journal. Your task is to conduct a deep, field-aware, construct-level critique of the manuscript ... Write in the tone of an experienced academic reviewer: precise, serious, analytically rigorous, and non-emotional. Avoid praise inflation. Avoid vague comments. Every critique must be grounded in reasoning.”

The original request required eight review blocks: (1) journal fit and genre compliance; (2) construct clarity and definition, including circularity and operationalizability; (3) novelty relative to prior scholarship and possible straw-person claims; (4) evidence and citation depth; (5) mechanistic specificity; (6) predictive and empirical viability; (7) page/line anchoring; and (8) an overall publication recommendation. The user additionally demanded ratings out of 10 across more than 20 categories.

Assistant verdict, faithfully summarized

> Overall: 7.0/10; major revision. The manuscript is an empirically developed, conceptually incremental original consumer-research paper, not a review article. It has a plausible integrated model and substantial survey sample, but its construct fidelity, mechanistic identification, integrated moderation transparency, fsQCA reporting, and robustness are not yet adequately established. The specific target of Journal of Retailing and Consumer Services was discussed as a plausible venue, not confirmed as an assigned journal.

Central positive contribution: perceived contemporary relevance remained positively associated with perceived cultural continuity after craftsmanship and authenticity appraisals were included. The moderation design potentially differentiated self-assessed knowledge from traditional value orientation. However, simultaneous significance is not proof of a distinctive mechanism or causal ordering.

3. Twenty-eight-dimensional editorial scoring rubric

Dimension	Score / 10
Journal scope alignment	8.2
Research significance	7.4
Problem formulation	8.0
Originality of research question	7.1
Theoretical novelty	6.5
Theoretical integration	7.0
Construct clarity	7.2
Construct operationalization	6.4
Conceptual discriminant validity	7.0
Literature breadth	7.7
Literature critical synthesis	7.0
Hypothesis development	7.4
Mechanistic specificity	6.0
Research design	6.8
Sample size and statistical power	8.3
Sampling and generalizability	6.6
Measurement-model reporting	7.8
PLS-SEM analysis	7.3
Moderation analysis	6.7
Serial indirect analysis	6.8
fsQCA rigor and transparency	5.6
Predictive validity	6.8
Statistical interpretation	7.7
Robustness and reproducibility	5.9
Results presentation	7.1
Discussion and implications	7.3
Writing and presentation quality	7.0
Ethics and research transparency	6.4

How to teach scoring: Explain the reason for each number using cited evidence. The 7.0 overall verdict was holistic rather than a mechanically computed average. Scores are expert editorial judgments, not calibrated probabilities of acceptance.

4. Complete construct-focused evaluation

Construct	Judgment	Issue that governs revision
TCA	7.0/10	Handmade skills, traditional technique, and artisan dedication may be formative facets rather than interchangeable reflective indicators.
PCA	7.5/10	Cultural typicality is not identical to genuine provenance or cultural legitimacy.
PCR	7.3/10	Current lifestyle fit is coherent but its scale source must be verified.
PCC	6.0/10	Historical memory, values continuity, cultural meaning, and traditional value are not all continuity indicators.
PK	7.8/10	Items measure subjective rather than factual knowledge.
TCO	6.5/10	Combines cultural pride, traditional values, family history, and preservation behavior.
PAR	8.1/10	Happy, positive, and good map relatively directly to positive affect.
BI	8.0/10	Measures buying intentions, not real purchasing.

Major comment 1 - PCC content validity (pp. 11 and 29; lines 391-400, Appendix A)

PCC1 assesses historical/cultural memory; PCC2 continuation of cultural values; PCC3 cultural meaning; PCC4 the value of traditional culture. Only PCC2 explicitly concerns persistence over time. A high reliability coefficient does not establish that a broad cultural-significance domain is a single well-defined temporal-continuity construct. A narrower definition, a defensible alternative label, or independent content-validation evidence is required.

Major comment 2 - TCA reflective specification (pp. 8-9; lines 295-325)

TCA combines related but potentially constitutive facets: perceived handmade skill, traditional production technique, and artisan effort. The reflective assumption must be justified on conceptual grounds, not inferred from Cronbach alpha or composite reliability. Where appropriate, test an alternative composite specification using existing item responses.

Major comment 3 - PCA typicality (p. 9; lines 326-336)

A culturally recognizable appearance may be typical without being historically authentic. Keep the boundary between respondent-inferred representational authenticity and externally verified provenance explicit. Conduct a content-validity review and distinguish typicality from authenticity.

Major comment 4 - TCO composition (p. 14; lines 508-527)

TCO mixes collective cultural pride, respect for tradition, traditional values, attachment to family history, and protection of family heritage. These are not necessarily interchangeable, which matters because TCO produces the strongest reported moderation coefficient. Test item-domain coherence and sensitivity.

Construct decision map

Variable	Current name	Working recommendation
TCA	Traditional Craftsmanship Appraisal	Keep; check measurement specification
PCA	Perceived Cultural Authenticity	Keep; explicitly limit to inferred authenticity
PCR	Perceived Contemporary Relevance	Keep; verify scale origin
PCC	Perceived Cultural Continuity	Keep only if content evidence supports it; otherwise consider Perceived Cultural Heritage Significance
PK	Product Knowledge	Clarify as Subjective Product Knowledge
TCO	Traditional Cultural Orientation	Keep provisionally; assess heterogeneity
PAR	Positive Affective Response	Keep
BI	Buying Intent	Keep

5. Theory, mechanism, empirical design, and statistics

Journal fit and contribution

The research fits consumer evaluation and heritage-product purchasing. It is original empirical research, not a systematic review. The identified gap - whether PCR remains significant after craftsmanship and authenticity are considered - is legitimate but bounded. The underlying literature already contains cue utilization, signaling, brand heritage, authenticity, cultural identity, value congruence, and general attitude-based explanations. A joint regression model does not independently adjudicate among these competing frameworks. Anchors: pp. 3-4, lines 81-104; p. 6, lines 200-206.

Major comment 5 - Cue process not observed (pp. 4-5, lines 105-118, 150-165)

The manuscript mentions provenance, production method, artisans, symbols, and functionality as cues, but does not manipulate or directly observe them. Its antecedents are self-reported appraisals, not experimentally assigned product information. Cue-utilization theory motivates the ordering but the model does not identify a cue-processing mechanism.

Major comment 6 - PCC to affect may be general evaluation (pp. 11-12, lines 409-449)

Value congruence is proposed but not directly measured. PCC items already contain favorable cultural evaluation, while PAR indicators directly solicit positive feelings. Their association may partly represent shared positivity rather than a unique value-congruence pathway. A competing-mechanism discussion and any supportable sensitivity models are needed.

Major comment 7 - Serial indirect path not causal (pp. 12, 21-22)

The reported serial standardized indirect estimates were 0.037 (TCA), 0.025 (PCA), and 0.040 (PCR). All are associations conditional on the ordering assumed in cross-sectional PLS-SEM. They do not establish which judgment temporally precedes which, or rule out omitted attitudes and reciprocal relationships.

Reported direct associations

Path	Reported β	Interpretation
TCA -> PCC	0.216	Positive conditional association
PCA -> PCC	0.146	Positive conditional association
PCR -> PCC	0.230	Positive conditional association
PCC -> PAR	0.454	Positive conditional association
PAR -> BI	0.380	Positive conditional association

The study reports R-squared values of 0.390 for PCC, 0.349 for PAR, and 0.144 for BI. Predictive Q-squared values include 0.074 for BI, 0.208 for PAR, and 0.360 for PCC. Such results are not evidence of strong individual-level purchase prediction.

Major comment 8 - Moderation interpretation (pp. 13-14, 21-22)

Interaction	β	f^2
PK x TCA -> PCC	0.220	0.041
PK x PCA -> PCC	0.152	0.024
PK x PCR -> PCC	0.174	0.031
TCO x PCC -> PAR	0.484	0.196

Report the integrated model with all lower-order terms, interaction formation and scaling, simple slopes, and confidence intervals. Evaluate the stability of TCO x PCC because both constructs contain culturally favorable evaluations. The PK interactions are positive but have small incremental effect sizes.

Major comment 9 - fsQCA needs re-execution (pp. 22-24)

The article gives ten configurations with overall solution coverage 0.85433 and consistency 0.824604, but configuration-level consistencies for S9-S10 appear as “NR.” Calibration uses standardized anchors +3 / 0 / -3, whose substantive appropriateness must be defended. Seven conditions permit 128 truth-table rows; frequency cutoff of four alone does not establish robustness. Reproduce all calibration, truth-table, minimization, necessity (including absences where relevant), directional expectations, and threshold sensitivity outputs. If original fsQCA output cannot be recovered, remove the section rather than filling missing statistics.

Major comment 10 - Out-of-sample prediction

Document cross-validation design, indicator-level comparisons, aggregation of RMSE and MAE, CVPAT benchmark choices, and the conventional linear-model benchmark where appropriate. BI performance is modest even if Q-squared is positive. Anchors: pp. 18-20, lines 653-660, 733-746.

Major comments 11-12 - Power and common reference object

N=496 is valuable, but an overall seven-predictor G*Power calculation does not specifically establish sensitivity for small interactions or serial indirect paths (pp. 15-16). The survey addresses an entire category of heritage goods, yet items alternate between “this cultural heritage product” and “cultural heritage products.” Clarify what each respondent was instructed to evaluate and assess product-type heterogeneity where feasible (pp. 15-17).

6. Evidence integrity, presentation, and editorial ruling

Major comment 13 - PCR scale source mismatch

The manuscript identifies Hu et al. (2024) as a perceived product-modernity scale source (pp. 10 and 17), but its listed Hu et al. citation is an article about Anglicisms, perceived risk, and appeal in Italian advertising (p. 32 references). That discrepancy requires primary-source verification and correction. Do not simply insert a convenient new citation without verifying original items.

Major comment 14 - Adaptation traceability

Construct provenance must connect each original scale, original wording, actual administered wording, pilot modifications, and conceptual interpretation. Evidence from related literature is not enough to establish equivalence after substantive adaptation (pp. 16-17).

Presentation and disclosure

Figure 1 on p. 16 still reads Perceived Cultural Sustainability (PCS), conflicting with the text and tables calling it Perceived Cultural Continuity (PCC). A discussion typo reads “This study The findings” (p. 24). The ethics and consent statements need complete, documented review status, collection dates, panel source, exclusions, and data availability. A reference dated 2027 in the bibliography should have its actual publication status checked for an October 2026 submission.

Original recommended editorial outcome

> “Major revision.” Do not reject merely because the study uses PLS-SEM or cross-sectional associations. The key scientific problems are construct fidelity, analytical transparency, mechanism differentiation, and the magnitude of the theoretical extension. Do not portray significant indirect effects as causal and do not claim the data demonstrate objective cultural preservation or actual purchasing behavior.

Top 11 revision actions from the review

- 1. Verify PCR scale provenance and correct the citation
- 2. Resolve PCC naming and content validity
- 3. Reproduce complete fsQCA or remove it
- 4. Publish the complete integrated moderation model and conditional slopes
- 5. Evaluate TCA measurement-specification alternatives
- 6. Test TCO composition and interaction stability
- 7. Clarify the questionnaire reference object, recruitment and screening
- 8. Make the novelty claim theoretically distinctive and bounded
- 9. Document predictive settings and interaction-specific sensitivity
- 10. Correct terminology in Figure 1 and edit prose
- 11. Complete ethics, consent, data, and reproducibility reporting

7. Conversation 2 - New data or renaming?

User follow-up

> “Find solutions if I needed to collect new data or rename the variable.”

Assistant answer - main decision

> Retain the current N = 496 dataset initially, preserve the main model, and address variable labeling, validation, robustness, and transparency before authorizing more data collection. New data are not automatically necessary to close every peer-review issue.

Variable	Recommended disposition	New data now?
TCA	Keep; test formative/composite robustness if justified	No
PCA	Keep; inferred cultural authenticity only	No
PCR	Keep; verify scale source and adaptation	No
PCC	Consider PCHS only if concept shift is acceptable	Not for narrower reinterpretation
PK	Subjective Product Knowledge (SPK)	No
TCO	Keep; audit subdomains	No initially
PAR	Keep	No
BI	Keep	No

Why PCC was singled out

The four PCC items refer to (1) heritage memory, (2) continuation of cultural values, (3) cultural meaning, and (4) the value of traditional culture. A candidate label was Perceived Cultural Heritage Significance (PCHS). That label could fit the broader content more closely, but it changes the substantive outcome from continuity over time to perceived cultural importance and meaning. It is not a find-and-replace operation.

What existing data can and cannot solve

Using the existing records, researchers may run robustness checks, inspect item behavior, estimate competing measurement specifications, recalculate conditional slopes, reconstruct fsQCA, and narrow their interpretations. However, no statistical test can make an absent temporal-continuity indicator appear in the administered questionnaire. If the goal is to measure actual cultural transmission, causal effects of product cues, objective consumer knowledge, or verified purchases, a new measurement design is necessary.

Research target	Data decision	Example design
Existing appraisal-path correlations	Use existing data	Robustness and conditional models
Item-content validity	Potential additional evidence	Independent expert ratings / validation sample
Actual preservation	New data	Longitudinal community and craft-transmission evidence

Research target	Data decision	Example design
Causal product-cue effect	New data	Randomized stimulus experiment
Objective product knowledge	New data	Knowledge assessment
Actual purchases	New data	Purchase records / follow-up
Modernity x authenticity tension	Try existing data first	Interaction and nonlinearity; experiment if needed

Decision priority: First audit PCC against the exact items, then audit adaptation sources and re-estimate robustness checks. Collect new observations only if the revised research claim demands properties that the existing instrument did not measure.

8. Conversation 3 - Can an adapted construct be renamed?

User follow-up

> “As per variable items and adaptation would it be fine to change variable name?”

Assistant answer - academically defensible conditions

Yes, a renamed construct may be appropriate when its new label better represents what the administered items actually measure. The name, conceptual definition, original scale, adapted wording, and results interpretation must stay aligned. Numerical reliability or HTMT results do not validate the conceptual meaning of a newly named construct.

Recommended names, preserved from the conversation

Original	Suggested	Decision
TCA	Traditional Craftsmanship Appraisal	Keep
PCA	Perceived Cultural Authenticity	Keep
PCR	Perceived Contemporary Relevance	Keep
PCC	Perceived Cultural Heritage Significance (PCHS)	Conditional; substantive theory change
PK	Subjective Product Knowledge (SPK)	Recommended clarification
TCO	Traditional Cultural Orientation	Keep pending dimensionality assessment
PAR	Positive Affective Response	Keep
BI	Buying Intent	Keep

The specific caution about PCHS

Under the continuity framing, the manuscript asks whether contemporary relevance signals that heritage culture remains continuous. Under PCHS, the question becomes whether contemporary relevance is associated with perceived cultural meaning and significance. Both could be useful, but they are not the same contribution. Renaming must trigger a full chain audit: construct definition -> hypotheses -> model diagram -> inference language -> table labels -> implications -> limitations.

The specific caution about PK

The four PK items concern how knowledgeable, familiar, comparatively knowledgeable, and confident a respondent feels. They do not examine factual correctness. “Subjective Product Knowledge” more accurately characterizes this item content. Keeping the code PK is possible if the full definition explicitly says self-assessed knowledge.

The specific caution about TCO

Avoid automatically changing TCO to “Traditional Cultural Attachment”: respect for tradition and endorsement of traditional values do not always mean attachment. Retaining TCO with a rigorous content and dimensionality justification is preferable to another unsupported relabeling.

9. Repeatable teaching workflow

Stage A - Obtain an evidence-grounded review

• Upload the complete manuscript and, if available, supplementary methods and tables.

• Ask for page-and-line anchored critical review, not generic “Q1 improvements.”

• Request 20+ dimensional scores with a reason and source location for each weak dimension.

• Require the reviewer to label observational findings, theoretical inference, and claims needing verification separately.

• Demand a prioritization based on threat to validity, not cosmetic importance.

Stage B - Convert criticism into falsifiable closure tasks

• Extract each reviewer claim as an actionable test or factual correction.

• Categorize: reporting only; existing-data reanalysis; independent citation/source check; new validation; new experiment.

• For statistical criticisms, recover and run the original code/data rather than inventing results.

• Retest conclusions after corrections; preserve negative or nonsignificant outcomes.

• Write a response-to-reviewers matrix: comment, action, evidence, manuscript location, status.

Stage C - Decide on name changes carefully

• Write a construct definition without seeing the variable name.

• Read each administered indicator and determine what it actually measures.

• Compare item meaning to source-scale wording and theory.

• Identify whether the proposed new label changes the dependent concept or only clarifies it.

• Either substantiate retention or revise all affected claims after renaming.

9. Reusable prompts for review-agent training

Prompt 1 - Full structured referee report

> Act as a senior reviewer for [JOURNAL]. Read the full manuscript, including measures, tables, figures, and references. Evaluate journal fit, research genre, construct definitions and operationalization, nearest literature and bounded novelty, citation evidence, mechanisms, rival explanations, design/identification, predictive and empirical viability, reproducibility, and reporting. Provide 25-30 category scores out of 10 with short reasons. For each major comment cite the exact page/line or section and state an actionable correction. Separate editorial problems from evidence limitations; do not invent data or treat statistical association as causation. End with a publication-readiness verdict.

Prompt 2 - Reviewer-comment closure

> Turn this report into a closure matrix. For every comment specify its evidence anchor, severity, the minimum acceptable correction, exact files/data needed, whether it can be solved with current observations, verification test, and closure status. Do not close any comment without observed supporting evidence. Prioritize threats to content validity and reproducibility.

Prompt 3 - Construct adaptation and renaming

> For each construct, compare the exact questionnaire items, cited original scale, source item wording, theoretical definition, and current manuscript interpretation. Decide KEEP / CLARIFY / RENAME / RE-MEASURE with specific justification. Identify any theory, hypotheses, tables, or results that would change after renaming. Do not confuse scale reliability with content validity.

Prompt 4 - New-data decision

> For each unresolved comment, state whether it requires (A) editorial clarification, (B) existing-data robustness analysis, (C) original-source verification, (D) independent validation, or (E) genuinely new experimental/longitudinal/behavioral data. Explain the smallest defensible study that would resolve each E-type problem.

Prompt 5 - Final independent audit

> Review the revised manuscript as a skeptical independent referee. Test whether every original major comment has been closed with evidence, whether variable terminology remains stable throughout figures, appendices and text, and whether all statistical statements accurately reflect reproduced outputs. Flag any remaining unverified item and do not assign high ratings merely because more sections were added.

10. Methodological integrity and teaching exercises

Five integrity rules

- A strong manuscript review must be anchored to actual manuscript language and tables.
- A revised construct label must match administered questions, not desired theoretical novelty.
- A cross-sectional structural path is not proof of a causal process.

• An unexplained fsQCA solution or missing source reference must be reproduced or qualified, never fabricated.

• A higher score is not a substitute for stronger evidence or a journal editorial decision.

Classroom exercises

• Ask participants to rate PCC content validity before revealing the reviewer conclusion.

• Ask them to rewrite the PCR -> PCC hypothesis under the proposed PCHS interpretation and identify what is lost.

• Give students an incomplete fsQCA table with S9/S10 marked NR; ask whether the analysis can be reported as complete.

• Have two reviewers independently score mechanistic specificity and reconcile differences using page/line evidence.

• Build a response-to-reviewers spreadsheet in which “closed” requires a reproducible artifact or textual evidence.

What was NOT proven in the conversation

The review did not independently execute the Beihang Referee ZIP, audit the raw N=496 records, verify every DOI, rerun the PLS-SEM model, or reproduce fsQCA results. Thus the ratings and recommendations are grounded in the manuscript as read, but they are not a code-verified research replication. A high-quality review agent should make this limitation visible.

Take-home lesson. The useful function of a review agent is not to make a paper sound more publishable. It is to produce testable criticisms, identify where the evidence supports or fails to support the claims, preserve an audit trail, and help the researcher decide which changes are feasible with existing data and which require additional measurement.